Diagonal asymptotics of square Hardinian arrays

A formula for every fixed parameter

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Abstract

Starting from the complementary-minor identity of Dougherty-Bliss and Kauers and the sum-of-squares reduction in the public $r = 2, 3$ note, we prove

$$H_r(n,n) \sim c_r\, 4^{rn} n^{-\binom{r}{2}}, \qquad c_r = \frac{2^{r(r-3)}}{3^{r^2}\pi^r} \prod_{j=1}^{r} \Gamma(j/2)^2,$$

for every fixed integer $r \geq 1$. In particular, $H_4(n,n) \sim 4 \cdot 256^n/(3^{16}\pi^2 n^6)$, in agreement with the published asymptotic table. The proof uses a Pfaffian sign identity, a finite-order local binomial expansion, alternating-polynomial cancellation, a Gaussian integral, and an explicit geometric tail bound. We also give a direct calculation of the four-row constant which does not require the Gaussian integral evaluation. The formula matches the published constants through $r = 5$, but its $r = 7$ specialization disproves the separate suggestion that every constant is a product of integer powers of $2, 3, \pi$. The results concern r fixed before n tends to infinity.

1 Problem, scope, and statement

For integers $r \geq 0$ and $n \geq 1$, let $H_r(n,n)$ count arrays $(a_{ij})_{1\leq i,j\leq n}$ with entries in $\mathbb{Z}_{\geq 0}$ such that

$$a_{11} = 0, \qquad a_{nn} = n - r - 1, \qquad |a_{ij} - (\max(i,j) - 1)| \leq r,$$

and each permitted step right, down, or diagonally down-right increases the entry by 0 or 1. These are the square arrays defined in [1]. The asymptotic table on page 12 of its 13-page journal version gives the fixed-parameter cases through $r = 5$. The source identity used below occurs in the proof of its Theorem 6, on page 9. This note extends the public $r = 2, 3$ treatment [2] to every fixed parameter.

For $n \leq r$, the prescribed endpoint is negative and the count is zero. For $n = r + 1$ the all-zero array is the unique array. In particular, $H_4(n,n) = 0$ for $n = 1, 2, 3, 4$ and $H_4(5,5) = 1$. These boundary cases do not change an asymptotic statement. All limits here are through integer n with r fixed first. The proof uses classical finite combinatorics and real analysis.

Theorem 1. *For each fixed integer $r \geq 1$, put $R = \binom{r}{2}$. Then*

$$H_r(n,n) \sim \frac{2^{r(r-3)}}{3^{r^2}\pi^r} \left(\prod_{j=1}^{r} \Gamma(j/2)^2 \right) 4^{rn} n^{-R}. \tag{1}$$

For $r = 0$, $H_0(n,n) = 1$ exactly. In particular,

$$\boxed{H_4(n,n) \sim \frac{4\,256^n}{3^{16}\pi^2 n^6}.} \tag{2}$$

The same theorem also proves the $r = 5$ row of the source table:

$$H_5(n,n) \sim \frac{16\,1024^n}{3^{23}\pi^2 n^{10}}.$$

The sum-of-squares identity and the probability interpretation in the next section are established starting material from the public note [2]. They are reproduced to fix all conventions and dependencies. No novelty is claimed for those identities, for Pfaffian algebra, for Stirling's expansion, or for the Gaussian Selberg integral.

2 Exact finite identities

Fix $n \geq r+1$ and put $N = n-2$. For a binomial coefficient outside its usual integer range, use the value zero. Rows and columns in every submatrix are in increasing order, and an empty determinant is 1. Write

$$M = \left[\binom{i+j}{i}\right]_{0\leq i,j\leq N}.$$

The counting identity from [1] is

$$H_r(n,n) = \sum_{\substack{I,J\subseteq\{0,\dots,N\}\\ |I|=|J|=r}} \det M_{I^c,J^c}. \tag{3}$$

Every subset occurs once; there is no factorial multiplicity.

Lemma 2 (Sum of squares and uniform bound)**.** *For $0 \leq u_1 < \cdots < u_r \leq N$, define*

$$W_r(u_1,\dots,u_r) = \sum_{0\leq i_1<\cdots<i_r\leq N} \det\left[\binom{u_a}{i_b}\right]_{a,b=1}^{r}, \tag{4}$$

$$w_r(u_1,\dots,u_r) = 2^{-\sum_a u_a} W_r(u_1,\dots,u_r). \tag{5}$$

Then

$$H_r(n,n) = \sum_{0\leq u_1<\cdots<u_r\leq N} W_r(u_1,\dots,u_r)^2. \tag{6}$$

If the X_a are independent with $X_a \sim \mathrm{Bin}(u_a, 1/2)$, then

$$w_r(u_1,\dots,u_r) = \mathbb{E}\left[\operatorname{sgn}\prod_{a<b}(X_b - X_a)\right], \qquad |w_r| \leq 1. \tag{7}$$

Here $\operatorname{sgn}(0) = 0$.

Proof. Set $P_{u,i} = \binom{u}{i}$ and $E = \mathrm{diag}((-1)^i)$. Vandermonde's binomial identity gives $M = PP^T$. Since P is unit lower triangular and $P^{-1} = EPE$, we have $\det M = 1$ and

$$M^{-1} = E(P^TP)E.$$

Jacobi's complementary-minor identity therefore gives

$$\det M_{I^c,J^c} = (-1)^{\sum I+\sum J}\det(M^{-1})_{J,I} = \det(P^TP)_{J,I}.$$

All matrices are finite and M is invertible, as required. Cauchy–Binet expands the last determinant as a sum over r-element row sets of P. Summing over I, J gives (6). Expanding (4) and sorting the distinct values of the independent X_a gives (7). A tie contributes zero to the sign and has no corresponding strictly increasing column set. Since the sign has absolute value at most one, the bound follows. □

For $u < v$, define

$$D(u,v) = \sum_{j=u}^{v-1} \binom{u+v}{j}, \qquad Q(u,v) = 2^{-u-v} D(u,v), \tag{8}$$

and extend D, Q skew-symmetrically, with diagonal entries zero. For independent $X \sim \mathrm{Bin}(u, 1/2)$ and $Y \sim \mathrm{Bin}(v, 1/2)$, $Y - X$ has the distribution of $Z - u$ for $Z \sim \mathrm{Bin}(u+v, 1/2)$. Symmetry about $(u+v)/2$ gives

$$\mathbb{E}\,\mathrm{sgn}(Y - X) = \mathbb{P}(Z > u) - \mathbb{P}(Z < u) = \mathbb{P}(u \le Z < v) = Q(u,v).$$

Lemma 3 (Exact Pfaffian reduction)**.** *Let $A_{ij} = Q(u_i, u_j)$ for $1 \le i, j \le r$. If r is even, then $w_r = \mathrm{Pf}\, A$. If r is odd, then*

$$w_r = \mathrm{Pf} \begin{pmatrix} A & \mathbf{1} \\ -\mathbf{1}^T & 0 \end{pmatrix}. \tag{9}$$

In particular,

$$w_4 = Q(u_1,u_2)Q(u_3,u_4) - Q(u_1,u_3)Q(u_2,u_4) + Q(u_1,u_4)Q(u_2,u_3). \tag{10}$$

Proof. For any even number of real numbers $x_1, \ldots, x_{2m}$,

$$\mathrm{Pf}[\mathrm{sgn}(x_j - x_i)]_{i,j=1}^{2m} = \mathrm{sgn} \prod_{i<j} (x_j - x_i). \tag{11}$$

If two values coincide, the corresponding rows and columns coincide and both sides are zero. If all values are distinct, permuting them into increasing order multiplies both sides by the same permutation sign. In increasing order the matrix has every entry above the diagonal equal to one. Its Pfaffian is one, by induction using the first-row expansion and $\sum_{j=2}^{2m} (-1)^j = 1$. This proves (11), including ties. For odd r, append a formal value $+\infty$, whose pairwise signs with the original variables are all one. In a Pfaffian matching each random variable appears in exactly one factor. Independence therefore factors the expectation of each product into pair expectations. Apply (7) and (8); the dummy factor in the odd case is deterministically one. □

3 A uniform local expansion to arbitrary fixed order

We need coefficient structure, not a central limit theorem without an error estimate. All constants in this section may depend on the fixed expansion order K and the fixed number C, but not on N, L, a, b in the specified ranges.

Lemma 4. *Fix $K \ge 0$ and $C > 0$. There are polynomials $P_k(a,b) \in \mathbb{Q}[a,b]$, $0 \le k \le K$, such that, uniformly for integer offsets $0 \le b < a \le L \le C \log N$,*

$$Q(N-a, N-b) = \frac{1}{\sqrt{\pi N}} \sum_{k=0}^{K} \frac{P_k(a,b)}{N^k} + O_{K,C}\left(\frac{(1+L)^{2K+3}}{N^{K+3/2}} \right). \tag{12}$$

Every P_k is antisymmetric in a, b, has total degree at most $2k+1$, and its homogeneous part of degree $2k+1$ is

$$[P_k(a,b)]_{2k+1} = \frac{(-1)^k (a-b)^{2k+1}}{4^k k!(2k+1)}. \tag{13}$$

In particular $P_0(a,b) = a - b$.

Proof. Put $d = a - b$, $s = a + b$, and $m = 2N - s$. The summation indices in (8) are $j = m/2 + t$, with

$$t = -d/2, -d/2+1, \ldots, d/2-1.$$

The values t may be half-integers; the factorial arguments are always integers. For sufficiently large N, all three factorial arguments $m, m/2+t, m/2-t$ are positive and comparable to N, uniformly in the window.

Here is the finite-order Stirling calculation including the degree information. Write $x = 2t/m$ and $c_h = B_{2h}/(2h(2h-1))$. For an integer J with $2J+1 \geq K+1$, the logarithmic Stirling formula with its positive-real-axis remainder bound [4] gives

$$\begin{aligned}
&\log\left(2^{-m}\binom{m}{m/2+t}\sqrt{\pi N}\right) \\
&= -\frac{1}{2}\log\left(1 - \frac{s}{2N}\right) - \frac{m}{2}\big((1+x)\log(1+x) + (1-x)\log(1-x)\big) - \frac{1}{2}\log(1-x^2) \\
&\quad + \sum_{h=1}^{J} c_h \left(m^{-(2h-1)} - (m/2+t)^{-(2h-1)} - (m/2-t)^{-(2h-1)}\right) + O_J(N^{-2J-1}).
\end{aligned} \tag{14}$$

One may increase J freely before this finite expansion. All arguments of Stirling's formula are positive reals, so its standard remainder bound applies; there is no large complex shift or unverified sector condition.

Taylor-expand (14) in N^{-1} through order K. The coefficient $\ell_j(s,t)$ at N^{-j} is an even polynomial in t of total degree at most $j+1$, and

$$\ell_1(s,t) = -t^2 + s/4 - 1/8.$$

To verify the degree bound explicitly, an entropy term is a multiple of $t^{2h} m^{-(2h-1)}$. Its N^{-j} coefficient is a multiple of $t^{2h} s^{j-2h+1}$, of degree $j+1$. The logarithmic prefactors give degree at most j, and the hth Stirling correction gives degree at most $j - 2h + 1$. The symmetry in t follows directly from (14).

The remainders of these Taylor expansions are uniform because $|s|/N, |t|/N \to 0$ uniformly and $s, |t| = O(\log N)$. More precisely they are $O_{K,C}((1+L)^{K+2} N^{-K-1})$ before exponentiation. Expanding the exponential now yields

$$2^{-m}\binom{m}{m/2+t} = \frac{1}{\sqrt{\pi N}}\left(\sum_{k=0}^{K} \frac{T_k(s,t)}{N^k} + O_{K,C}\left(\frac{(1+L)^{2K+2}}{N^{K+1}}\right)\right). \tag{15}$$

Indeed, the whole logarithm is $O((1+L)^2/N)$, and a Taylor remainder of order $K+1$ in its exponential has precisely the displayed bound. The polynomial T_k is even in t and has total degree at most $2k$. Its degree-$2k$ part is uniquely $(-t^2)^k/k!$: an ℓ_j with $j > 1$ has degree $j + 1 < 2j$, and the other terms of ℓ_1 have degree below two.

For each $h \geq 0$, Faulhaber's identity gives

$$\sum_{v=0}^{d-1}(v-d/2)^{2h} = \frac{B_{2h+1}(d/2) - B_{2h+1}(-d/2)}{2h+1}. \tag{16}$$

This is an odd polynomial in d of degree at most $2h+1$, with leading term $d^{2h+1}/(4^h(2h+1))$. Summing each monomial of $T_k(s,t)$ over this grid therefore gives an odd polynomial in d, with s unchanged, of total degree at most $2k+1$. It is antisymmetric in a, b, since interchanging them fixes s and negates d. The unique top-degree term gives (13). Finally there are $d \leq L$ summands, so the error in (15) becomes the error in (12). This also proves uniformity across both possible grid parities. □

4 Why every lower asymptotic order cancels

For decreasing offsets $a_1 > \cdots > a_r \geq 0$, define

$$\Delta(a) = \prod_{1 \leq i < j \leq r}(a_i - a_j) > 0.$$

It has degree $R = \binom{r}{2}$. For integer distinct offsets, $\Delta(a) \geq 1$.

Proposition 5. *Fix $r \geq 2$ and $C > 0$. There is a constant κ_r such that, uniformly for $0 \leq a_r < \cdots < a_1 \leq L \leq C \log N$,*

$$w_r(N-a_1, \ldots, N-a_r) = \kappa_r \Delta(a) N^{-R/2} + O_{r,C}\left((1+L)^{R+2} N^{-R/2-1}\right). \tag{17}$$

The constant is the coefficient of $t^R \Delta(a)$ in the even or augmented Pfaffian with ordinary entries $\operatorname{erf}(t(a_i - a_j)/2)$.

Proof. Let $m = \lfloor r/2 \rfloor$ and set

$$K_r = \frac{R-m}{2} = \begin{cases} m(m-1), & r = 2m, \\ m^2, & r = 2m+1. \end{cases}$$

This is a nonnegative integer. Every matching in Lemma 3 has exactly m ordinary entries. In the odd case its one remaining entry is the constant one associated with the dummy index.

Insert Lemma 4 with $K = K_r$ into the Pfaffian and remove the common factor $(\pi N)^{-m/2}$. Its coefficient at N^{-k} is an alternating polynomial in $a_1, \ldots, a_r$ of total degree at most $m + 2k$. This follows because a simultaneous permutation of ordinary rows and columns changes the Pfaffian by the permutation sign. Every alternating polynomial over $\mathbb{Q}$ is divisible by $\Delta(a)$: it vanishes when $a_i = a_j$, hence each linear factor $a_i - a_j$ divides it, and these distinct prime factors have a product dividing it. Thus all coefficients with $k < K_r$ vanish, since their degree is below R. The coefficient at $k = K_r$ is $\lambda_r \Delta(a)$ for a constant λ_r.

At that first possible order, only the top homogeneous parts (13) can contribute. These are exactly the coefficients in

$$\operatorname{erf}\left(\frac{td}{2}\right) = \frac{1}{\sqrt{\pi}} \sum_{k \geq 0} \frac{(-1)^k d^{2k+1} t^{2k+1}}{4^k k!(2k+1)}.$$

Therefore $\kappa_r = \pi^{-m/2} \lambda_r$ is characterized as stated.

For clarity, the remainders are controlled before using any cancellation. Each actual or truncated ordinary entry is $O_{r,C}((1+L)N^{-1/2})$. Replacing one entry by the remainder from (12) gives, in a product of m entries,

$$O_{r,C}\left((1+L)^{2K_r+3+m-1}N^{-K_r-3/2-(m-1)/2}\right) = O_{r,C}\left((1+L)^{R+2}N^{-R/2-1}\right).$$

The discarded terms of total index $k > K_r$ in the product of truncated polynomials satisfy the same bound, since $(1+L)^2/N \to 0$. There are only finitely many pairings for fixed r. This proves (17). □

5 Identification of the universal constant

For completeness, set $\kappa_1 = 1$, as given by the augmented Pfaffian when $r = 1$.

Lemma 6. *For each $r \geq 1$,*

$$\kappa_r = \frac{\prod_{j=1}^{r}\Gamma(j/2)}{\pi^{r/2}\prod_{j=0}^{r-1}j!} > 0. \tag{18}$$

Proof. Fix any strictly decreasing real tuple $a_1 > \cdots > a_r$, and let the independent variables Y_i be normal with mean $-ta_i$ and variance one. Their pairwise sign expectations are $\operatorname{erf}(t(a_i - a_j)/2)$. Consequently the Gaussian version of Lemma 3 identifies its Pfaffian as

$$F_a(t) = \mathbb{E}\operatorname{sgn}\prod_{i<j}(Y_j - Y_i).$$

Let $\phi(x) = (2\pi)^{-1/2}e^{-x^2/2}$. Splitting $\mathbb{R}^r$ into its ordering chambers gives

$$\begin{aligned} F_a(t) &= \int_{x_1<\cdots<x_r} \det[\phi(x_j + ta_i)]_{i,j=1}^{r}\,dx \\ &= e^{-t^2\sum_i a_i^2/2}\int_{x_1<\cdots<x_r} \det[e^{-ta_ix_j}]_{i,j=1}^{r}\prod_j \phi(x_j)\,dx. \end{aligned} \tag{19}$$

In the determinant of exponentials the first nonzero Taylor coefficient is

$$[t^R]\det[e^{-ta_ix_j}] = \frac{\Delta(a)\prod_{i<j}(x_j - x_i)}{\prod_{j=0}^{r-1}j!}.$$

To verify this, expand each exponential in monomials. Distinct powers are required in a nonzero determinant; their smallest possible sum is $0 + 1 + \cdots + (r-1) = R$. The corresponding two determinants are the Vandermonde determinants in $-a_i$ and in x_j; the first is $\Delta(a)$.

Differentiation through the integrals to this finite order is valid. For $|t| \leq 1$ and fixed a, every derivative through order $R+1$ of the Gaussian densities is bounded by a polynomial in $|x|$ times $C_a e^{-\sum_j x_j^2/4}$, which is integrable. Thus if $Z_1, \ldots, Z_r$ are independent standard normals,

$$[t^R]F_a(t) = \frac{\Delta(a)}{r!\prod_{j=0}^{r-1}j!}\mathbb{E}\left|\prod_{i<j}(Z_j - Z_i)\right|. \tag{20}$$

The factor $1/r!$ is the passage from one ordering chamber to all chambers.

The Gaussian Selberg (Mehta) integral at parameter $\gamma = 1/2$ is

$$\mathbb{E}\left|\prod_{i<j}(Z_j - Z_i)\right| = \prod_{j=1}^{r}\frac{\Gamma(1+j/2)}{\Gamma(3/2)}. \tag{21}$$

This is the normalization in [3, eqs. (1.5)–(1.8)]. Its hypotheses hold: r is a positive integer, $\gamma = 1/2$ is in the positive-parameter domain, the integrand has no nonintegrable collision singularities, and its tails are a fixed polynomial times a Gaussian. Combining (20) and (21), then using $\Gamma(1 + j/2) = (j/2)\Gamma(j/2)$ and $\Gamma(3/2) = \sqrt{\pi}/2$, proves (18). All gamma arguments are positive, so $\kappa_r > 0$. □

6 The weighted discrete Vandermonde sum

Lemma 7. *For a positive integer r and $0 < q < 1$, define*

$$Z_r(q) = \sum_{0 \le b_1 < \cdots < b_r} q^{\sum_i b_i} \prod_{i<j} (b_j - b_i)^2.$$

Then

$$Z_r(q) = \frac{q^{\binom{r}{2}} \prod_{j=0}^{r-1} (j!)^2}{(1-q)^{r^2}}. \tag{22}$$

Proof. Let $\mu_k(q) = \sum_{b \ge 0} b^k q^b$, where $0^0 = 1$. All these moments converge. Cauchy–Binet applied first to a finite weighted Vandermonde matrix, then taking its number of columns to infinity, gives

$$Z_r(q) = \det[\mu_{i+j}(q)]_{0 \le i,j < r}.$$

The passage to infinity is legitimate because each moment converges and a finite determinant is a polynomial in its entries.

For an elementary evaluation of this determinant, define polynomials $M_j(x)$ by

$$\sum_{j \ge 0} M_j(x) z^j = \frac{(1 - z/q)^x}{(1-z)^{x+1}}, \qquad x \in \mathbb{Z}_{\ge 0}.$$

The degree of M_j is j and its leading coefficient is $(1 - 1/q)^j / j!$. Summing a geometric series gives

$$\sum_{x \ge 0} q^x \frac{(1 - z/q)^x}{(1-z)^{x+1}} \frac{(1 - w/q)^x}{(1-w)^{x+1}} = \frac{1}{(1-z)(1-w) - q(1 - z/q)(1 - w/q)} = \frac{1}{(1-q)(1 - zw/q)}.$$

For z, w in a sufficiently small neighborhood of zero the geometric series converges absolutely and uniformly on smaller neighborhoods, since its ratio is $q < 1$ at zero. Coefficient comparison is therefore valid and yields

$$\sum_{x \ge 0} q^x M_j(x) M_k(x) = \frac{q^{-j}}{1-q} \, \delta_{jk}.$$

After making the polynomials monic, their squared norms are

$$h_j = \frac{(j!)^2 q^j}{(1-q)^{2j+1}}.$$

The change from monomials to monic polynomials is unit triangular, so the moment determinant is $\prod_{j=0}^{r-1} h_j$. This is (22). □

7 Tail control and completion of the theorem

Proof of Theorem 1. The $r = 0$ case is immediate from the distance constraint. For $r = 1$, the sum-of-squares identity gives

$$H_1(n, n) = \sum_{u=0}^{n-2} 4^u = \frac{4^{n-1} - 1}{3},$$

with the empty sum at $n = 1$. This has the stated constant $1/12$.

Fix now $r \geq 2$. Replacing row indices by decreasing offsets in (6) gives

$$\frac{N^R H_r(n, n)}{4^{rN}} = \sum_{N \geq a_1 > \cdots > a_r \geq 0} 4^{-\sum_i a_i} N^R w_r(N - a_1, \ldots, N - a_r)^2. \tag{23}$$

Choose

$$L = \lceil (R + 2) \log_4 N \rceil .$$

The window $L = O_r(\log N)$ is covered by Proposition 5 for a fixed C depending only on r. Outside it, the global bound $|w_r| \leq 1$ gives

$$\begin{aligned} &N^R \sum_{\max_i a_i > L} 4^{-\sum_i a_i} \\ &\quad \leq r N^R \left(\sum_{a > L} 4^{-a} \right) \left(\sum_{a \geq 0} 4^{-a} \right)^{r-1} = \frac{r}{3} \left(\frac{4}{3} \right)^{r-1} N^R 4^{-L} = O_r(N^{-2}). \end{aligned} \tag{24}$$

Here the sum was enlarged to all nonnegative integer tuples, so it also bounds all allowable decreasing tuples.

Inside the window, $\Delta(a) \geq 1$ and $\kappa_r > 0$. The relative error in (17) is consequently $O_r((1 + L)^{R+2}/N) = o(1)$ uniformly. Squaring gives

$$N^R w_r(N - a_1, \ldots, N - a_r)^2 = \kappa_r^2 \Delta(a)^2 (1 + o(1))$$

uniformly there. The weighted main sum converges to $Z_r(1/4)$, which is finite by Lemma 7. The uniform relative error therefore contributes $o(1)$ to the sum. Combining this with (24) yields

$$H_r(n, n) \sim 4^{rN} N^{-R} \kappa_r^2 Z_r(1/4). \tag{25}$$

No exchange of a merely pointwise limit with an uncontrolled infinite sum is used.

Finally $N = n - 2$, and

$$\begin{aligned} c_r &= 4^{-2r} \kappa_r^2 Z_r(1/4) \\ &= 4^{-2r} \frac{\prod_{j=1}^{r} \Gamma(j/2)^2}{\pi^r \prod_{j=0}^{r-1} (j!)^2} \frac{4^{r^2 - R} \prod_{j=0}^{r-1} (j!)^2}{3^{r^2}} \\ &= \frac{2^{r(r-3)}}{3^{r^2} \pi^r} \prod_{j=1}^{r} \Gamma(j/2)^2. \end{aligned}$$

We used $N/n \to 1$ and $r^2 - R - 2r = r(r - 3)/2$. This proves (1). For $r = 4$, the gamma product before squaring is $\Gamma(1/2)\Gamma(1)\Gamma(3/2)\Gamma(2) = \pi/2$, giving $c_4 = 4/(3^{16}\pi^2)$ exactly. □

8 A direct four-row calculation

This section gives an independent, elementary extraction of the required four-row coefficient. In particular, the $r = 4$ conclusion does not depend on the general Gaussian integral evaluation in Lemma 6.

Put $d = a - b$ and $s = a + b$. Lemma 4 through $K = 2$ reads

$$Q(N-a, N-b) = \frac{d}{\sqrt{\pi N}}\left(1 + \frac{A_1(s,d)}{N} + \frac{A_2(s,d)}{N^2}\right) + O\left((1+L)^7 N^{-7/2}\right), \tag{26}$$

$$A_1(s,d) = \frac{s}{4} - \frac{d^2}{12} - \frac{7}{24},$$

$$A_2(s,d) = \frac{d^4}{160} - \frac{sd^2}{16} + \frac{3s^2}{32} + \frac{3d^2}{32} - \frac{7s}{32} + \frac{61}{640}.$$

For example the mass at $j = m/2 + t$ has first two coefficients

$$T_1(s,t) = s/4 - t^2 - 1/8,$$

$$T_2(s,t) = 3s^2/32 - 3st^2/4 - 3s/32 + t^4/2 + 5t^2/8 + 1/128.$$

Summing these and using

$$\sum t^2 = \frac{d(d^2+2)}{12}, \qquad \sum t^4 = \frac{d(3d^4 + 20d^2 - 8)}{240}$$

gives (26).

Insert it into (10). After the common factor $(\pi N)^{-1}$, the coefficients of N^0 and N^{-1} are alternating polynomials of degrees at most two and four. Both vanish. The coefficient of N^{-2} is an alternating polynomial of degree at most six, hence a constant times $\Delta(a_1, a_2, a_3, a_4)$. Only the leading terms

$$g(d) = d - \frac{d^3}{12}\varepsilon + \frac{d^5}{160}\varepsilon^2$$

are needed to determine that constant. At $(a_1, a_2, a_3, a_4) = (3, 2, 1, 0)$,

$$[\varepsilon^2](g(1)^2 - g(2)^2 + g(3)g(1)) = \frac{1}{2}, \qquad \Delta(3,2,1,0) = 12.$$

Thus the constant is $1/24$, and

$$w_4(N-a_1, \ldots, N-a_4) = \frac{\Delta(a)}{24\pi N^3} + O\left((1+L)^8 N^{-4}\right). \tag{27}$$

The remaining constant is

$$Z_4(q) = \frac{144q^6}{(1-q)^{16}}, \qquad Z_4(1/4) = \frac{144 \cdot 4^{10}}{3^{16}}.$$

Equations (25) and (27) give

$$H_4(n,n) \sim \frac{256^N}{576\pi^2 N^6} \frac{144 \cdot 4^{10}}{3^{16}}.$$

Replacing N by $n - 2$ contributes the essential factor $256^{-2} = 4^{-8}$. The coefficient becomes

$$\frac{144 \cdot 4^{10}}{576 \cdot 4^8 \cdot 3^{16}\pi^2} = \frac{4}{3^{16}\pi^2},$$

which independently verifies the exact target constant.

9 Constants, source coverage and limits of scope

Theorem 1 holds for each fixed integer $r \geq 1$, with $H_0(n, n) = 1$ separately. It makes no uniform assertion for r growing with n and no assertion about joint rectangular limits. The exponent and constants agree with all rows through $r = 5$ in [1, p. 12]:

r	1	2	3	4	5
c_r	$\frac{1}{12}$	$\frac{1}{2^2 3^4 \pi}$	$\frac{1}{2^2 3^9 \pi}$	$\frac{2^2}{3^{16} \pi^2}$	$\frac{2^4}{3^{23} \pi^2}$

The formula also gives

$$c_6 = \frac{2^{14}}{3^{34} \pi^3}, \qquad c_7 = \frac{2^{18}\, 5^2}{3^{45} \pi^3}.$$

Thus the source's further suggestion that all constants use only powers of $2, 3, \pi$ cannot hold with integer exponents for every r. The factor 5^2 at $r = 7$ cannot be absorbed into those powers, by the transcendence of π and unique prime factorization. This concerns a separate informal general pattern: it does not contradict the displayed rows of the source table or the fixed-parameter asymptotic scale. Allowing arbitrary real exponents would remove the intended prime-factor restriction.

Related rectangular work. Dougherty-Bliss and Spahn's published four-page ISSAC poster abstract *Rectangular Hardinian Arrays* [5], in ACM Communications in Computer Algebra, is available and has been read for this comparison. For fixed r and fixed width k, its theorem makes $H_r(n, k)$ a polynomial of degree r in n for sufficiently large n, using a finite transfer matrix. Its hypotheses keep k fixed; they do not permit substituting $k = n$ or give a uniform threshold and asymptotic estimate in growing k. That theorem therefore does not supply the square-array result proved here. This is a comparison of specified sources, not certification of worldwide publication priority.

Proof dependencies and finite evidence. The original complementary-minor count is an external combinatorial input from [1]. The square-sum and sign-expectation reductions from the public note [2] are rederived here with all conventions fixed. Jacobi's identity, Cauchy–Binet, Stirling's formula with its stated remainder, Faulhaber's identity and the Mehta integral are used explicitly at their respective steps. In particular, the all-r constant uses the standard Gaussian integral [3], whereas the four-row calculation extracts its coefficient directly.

No finite computation is needed to conclude the theorem. Earlier records contain both an author-reported exact-data run and separate later finite checks; the distribution's reproducibility note distinguishes these records from this manuscript preparation, during which no mathematical program was run. File hashes identify versions and do not themselves certify execution. None of these finite checks replaces the arbitrary-order expansion, Vandermonde cancellation, integral normalization or full tail estimate.

Version and verification status. This is the fixed-parameter extension volume of the public $r = 2, 3$ note [2]; the earlier manuscript, code and data remain unchanged historical material. Prepared with AI assistance. No independent human peer-review, proof-assistant certification or worldwide-priority claim is made.

References

[1] R. Dougherty-Bliss and M. Kauers, *Hardinian Arrays*, Electronic Journal of Combinatorics **31**(2) (2024), P2.9, 13 pages. doi:10.37236/12358. Theorem 6, p. 9; asymptotic table, p. 12.

[2] Aurantium, *Two asymptotic formulas for square Hardinian arrays*, public research note, version 0.1.0 (2026). Public version and source archive. Fixed source commit 8bf9f116a7d49c6b49ecfec9d30d14a2afc4502e. Section 2 derives the general square-sum and probability identities; the asymptotic conclusions of that volume cover $r = 2, 3$.

[3] P. J. Forrester and S. O. Warnaar, *The importance of the Selberg integral*, Bulletin of the American Mathematical Society (N.S.) **45** (2008), 489–534. arXiv:0710.3981. The Gaussian integral normalization and evaluation are (1.5)–(1.8).

[4] NIST Digital Library of Mathematical Functions, §5.11(i)–(ii), logarithmic Stirling expansion and positive-real-axis remainder bounds.

[5] R. Dougherty-Bliss and G. Spahn, *Rectangular Hardinian Arrays*, ACM Communications in Computer Algebra **58**(3), issue 229 (September 2024), 81–84, ISSAC 2024 poster abstracts. doi:10.1145/3717582.3717589.